

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack platform for curating and preprocessing **ML training datasets** (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Implemented backend security controls, AWS KMS encryption, auth hardening, and rate limiting, remediating **20+ VAPT findings**.
- Served as primary technical POC and piloted a COO-sponsored GitHub Copilot experiment, using AI-assisted development to accelerate tech-debt and migration work.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Shipped a production feature deploying a **4-bit quantized Llama-7B** model via llama.cpp with async queued generation and a content-safety filter, built end to end with no ML mentor.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) and migrated core repositories from JavaScript to TypeScript.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform independently end to end; live **3+ years**.

Projects

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built an on-device **multimodal** speech and vision system integrating MediaPipe hand tracking, gesture classification, **whisper.cpp ASR**, and local **TTS** into one offline sign-and-speech pipeline (FastAPI backend, React frontend) with async APIs.

Refusal Decay – LLM Safety / Mechanistic Interpretability (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Ran a mechanistic-interpretability study of transformer refusal behavior in Llama-3.1-8B-Instruct (**PyTorch**), extracting the residual-stream refusal direction via difference-in-means (refusal **0.92**→**0.32** at $k=3$) with causal patching, bootstrap 95% CIs, and McNemar's test.

ngvi-curvature-variance – Natural-Gradient VI Study (GitHub) Python, PyTorch, NumPy

- Ran a matched-compute study of natural-gradient variational inference vs Adam on ill-conditioned posteriors (**PyTorch**, black-box VI), instrumenting Fisher geometry and gradient variance across experiments.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, SQL, Bash, JavaScript, TypeScript

Deep Learning: PyTorch, JAX, Hugging Face Transformers, NumPy, transformers, evaluation

Generative: diffusion (DDPM), normalizing flows, variational inference, ELBO, natural-gradient methods

Speech & Vision: ASR (whisper.cpp), TTS, MediaPipe, CNNs, transfer learning, multimodal pipelines

Cloud & Tools: AWS, GCP, Azure, Docker, Git, Cursor / Copilot

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.